MD Tanvir Ahammed Hridoy

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RESEARCH INTEREST

Computer Science researcher and M.Sc. student with a first-author publication in *Social Network Analysis and Mining* (Springer Nature). Research experience in machine learning, NLP, web scraping, ensemble learning, deep learning, and predictive modeling. Currently interested in extending this foundation toward AI-driven cybersecurity, particularly trustworthy intrusion detection and secure software design, developing machine-learning methods that remain reliable and interpretable under real-world constraints.

EDUCATION

Jagannath University — **September 2024 – October 2026 (Expected)**
M.Sc. in Computer Science – ***In Progress*** — *Dhaka, Bangladesh*

Jagannath University — **January 2019 – July 2024**
B.Sc. in Computer Science & Engineering – ***CGPA: 3.59/4.00*** — *Dhaka, Bangladesh*

Holy Land College — **June 2016 – June 2018**
Higher Secondary School Certificate (Science) – ***GPA: 4.92/5.00*** — *Dinajpur, Bangladesh*

RESEARCH EXPERIENCE

Undergraduate Thesis: Detection of MDD | Springer Nature — **June 2023 – April 2024**

- **First-author** publication: "Leveraging Web Scraping and Stacking Ensemble Machine Learning Techniques to Enhance Detection of Major Depressive Disorder from Social Media Posts," *Social Network Analysis and Mining*, Springer Nature.
- Utilized PRAW API for large-scale data collection from Reddit; built advanced NLP pipelines incorporating sentiment analysis, feature extraction, and hyperparameter optimization.
- Designed a Stacking Ensemble framework integrating multiple classifiers and deep learning models, achieving high-accuracy predictive diagnostics for mental health monitoring at scale.
- **Core Techniques:** Web Scraping, NLP, Deep Learning, Stacking Ensemble, Python, Scikit-Learn, TensorFlow.

TECHNICAL SKILLS

AI & Machine Learning: Deep Learning, NLP, Stacking Ensemble, Transformers, TensorFlow, Scikit-learn
LLMs & Advanced AI: Hugging Face Transformers, SHAP (XAI)
Programming Languages: Python, Swift, C++, C#, Java
Security & Systems: Secure API Design, REST Security, Data Integrity
App Development: iOS (SwiftUI), Flutter, Dart, CocoaPods, REST APIs
Cloud & DevOps: Git, GitHub, Docker, Kubernetes, CI/CD Pipelines
Problem Solving: 500+ problems solved on LeetCode, Codeforces, UVA, and URI

WORK EXPERIENCE

Bdjobs.com Limited — **August 2024 – Present**
Application Software Developer (iOS) — *Dhaka, Bangladesh*

- Developed and maintained a large-scale iOS application serving thousands of daily users, implementing secure REST API integration with token-based authentication and encrypted local data handling (SQLite).
- Collaborated with cross-functional teams on system architecture decisions, applying software security best practices for API communication and user data protection.

Jagannath University (CSE) & EDGE — **March 2024 – February 2025**
Python with Machine Learning Trainee — *Dhaka, Bangladesh*

- Developed practical skills in Python-based machine learning pipelines including data preprocessing, model training, evaluation, and deployment for real-world AI applications.
- Gained hands-on experience building and evaluating deep learning models using TensorFlow and Scikit-learn for classification and regression tasks.

Prime Tech Solutions Ltd. & IIT, University of Dhaka **September 2023 – December 2023**
Cloud Computing Trainee *Dhaka, Bangladesh*

- Gained hands-on experience implementing CI/CD pipelines for continuous integration and deployment workflows.
- Deployed and managed applications using Docker and Kubernetes, ensuring scalable and efficient orchestration in cloud environments.

PROJECTS

1. Bdjobs iOS App | App Store Link **August 2024 – Present**

- Production-scale job portal iOS application serving active users nationwide, with emphasis on secure data handling, API security, and system reliability.
- Key contributions: Job Details, Job List, Employer Messaging, Job Fair Enhancements, Early Access Listings, Renew/Repurchase/Upgrade Flows, and Plans & Pricing.
- **Tech Stack:** Swift, SQLite, CocoaPods, REST APIs, Google Firebase, Google AdSense.

2. IoT-Based Intelligent Health Monitoring System | GitHub **January 2023 – May 2023**

- Designed an intelligent embedded system for real-time health monitoring, integrating ECG, pulse, and thermal sensors with ESP8266-driven IoT connectivity.
- Developed a robust data pipeline for physiological signal acquisition and analysis, bridging hardware-level sensing with cloud-based predictive diagnostics to enable remote medical assistance.
- **Tech Stack:** Arduino UNO, ESP8266 (Wi-Fi Module), AD8232 (ECG Sensor), XD58C (Pulse Sensor).

3. Detecting Counterfeit Banknotes | GitHub | Demo Video **June 2023 – December 2023**

- Engineered a counterfeit currency detection system for Bangladeshi banknotes utilizing digital image processing (DIP) techniques, including image acquisition and preprocessing.
- Implemented contour analysis and watermark detection algorithms to accurately authenticate security elements and isolate embedded features within the notes.
- **Tech Stack:** Python, OpenCV, NumPy.

4. JnU Canteen App & Website | App Link | Web Link **January 2024 – June 2024**

- IoT-inspired smart campus system integrating meal ordering, bKash payment gateway, and QR code scanning for real-time order tracking and automated menu management.
- **Tech Stack (App):** Flutter, Dart, Firebase; **Tech Stack (Website):** HTML, CSS, JavaScript, PHP, MySQL, Ajax.

ACHIEVEMENTS

- **University Merit Scholarship:** Awarded by Jagannath University for maintaining outstanding academic performance throughout the B.Sc. program.
- **JNU Project Showcasing Competition:** Secured **3rd Place** (2023) and **3rd Place** (2022) as a member of Team_Fixit for excellence in software innovation.
- **Programming Contests:** Ranked **7th** (2022) and **5th** (2023) in the Intra-University Programming Contests (Handle: hridoy65). Participated in **ICPC** and various national hackathons.
- **Orange Corners BD Ideation Challenge 3.0:** Recognized as a **Top 10 Finalist** (Team AgriVillage) in the global entrepreneurship and ideation initiative.
- **JNU Power Presenter 2023:** Achieved **2nd Runner-Up** in the university-wide presentation and communication skills competition.

EXTRACURRICULAR

Jagannath University Debating Society (JnUDS) **July 2021 – May 2024**
Department Representative *Dhaka, Bangladesh*

- Represented the Department of CSE for three consecutive years.
- Coordinated activities and guided new members, contributing to the team becoming finalists twice.

YouTube Channel: "TechBD For You" **September 2021 – Present**
Technical Content Creator *Dhaka, Bangladesh*

- Produce educational content on cybersecurity, AI/ML concepts, database management, and software systems to support student learning and technical literacy.
- Demonstrate research-oriented university projects through practical walkthroughs and real-world implementation.